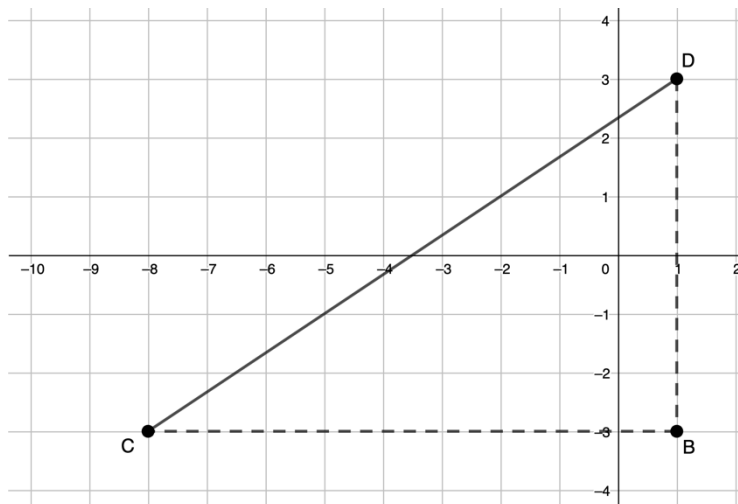


Geometry
Unit 13
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Find the length of segment CD using the Pythagorean Theorem. If necessary, round to the nearest thousandth.



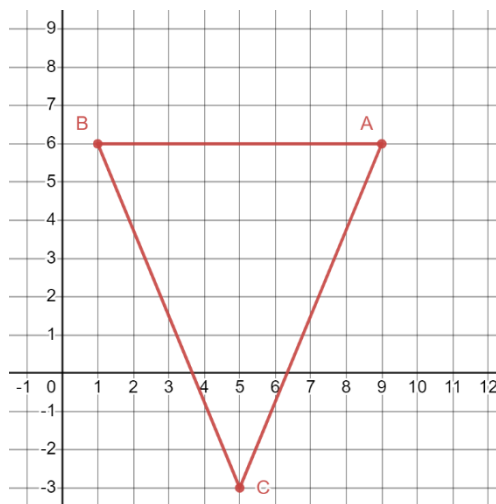
$$\begin{aligned} &C(-8, -3), D(1, 3) \\ &\sqrt{(-8 - 1)^2 + (-3 - 3)^2} \\ &= \sqrt{(-9)^2 + (-6)^2} \\ &= \sqrt{81 + 36} = \sqrt{117} \\ &= 3\sqrt{13} \approx 10.817 \end{aligned}$$

$$\overline{CD} = 10.817 \text{ units}$$

Use the following information for questions 2-6.

$\triangle ABC$ has vertices at $A(9,6)$, $B(1,6)$, and $C(5,-3)$.

2. Graph $\triangle ABC$ on the coordinate plane.



Find the length of each segment. If necessary, round to the nearest tenth.

	Line Segment	Length of Line Segment
3.	$\overline{AB}$	$9 - 1 = 8 \text{ units}$
4.	$\overline{BC}$	$\sqrt{(5 - 1)^2 + (-3 - 6)^2} = \sqrt{97} \approx 9.8 \text{ units}$
5.	$\overline{AC}$	$\sqrt{(5 - 9)^2 + (-3 - 6)^2} = \sqrt{97} \approx 9.8 \text{ units}$

6. Complete the statement.

$\triangle ABC$ is classified as a(n)

- ☐ equilateral
- ☒ isosceles
- ☐ scalene

triangle because $\triangle ABC$ has

- ☐ no
- ☒ two
- ☐ three

equal side lengths.

Use the following information for questions 7-10.

$\triangle TUV$ has vertices at $T(1, -8)$, $U(8, -1)$, $V(4, -1)$.

Find the length of each segment. If necessary, round to the nearest tenth.

	Line Segment	Length of Line Segment
7.	$\overline{TU}$	$\sqrt{(8-1)^2 + (-1-(-8))^2} = \sqrt{98} \approx 9.9$ units
8.	$\overline{UV}$	$8-4 = 4$ units
9.	$\overline{TV}$	$\sqrt{(4-1)^2 + (-1-(-8))^2} = \sqrt{58} \approx 7.6$ units

10. Complete the statement.

$\triangle TUV$ is classified as a(n)

- ☐ equilateral
- ☐ isosceles
- ☒ scalene

triangle because $\triangle TUV$ has

- ☒ no
- ☐ two
- ☐ three

equal side lengths.